

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Product Engineer”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

PAVITHRA R - 1DS20AI038

Conducted at



Subex Limited

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**Product Engineer**” carried out by Ms. **PAVITHRA R [IDS20AI038]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)


Signature of Reporting
Head
Upendra K B
Team Lead
Subex Limited

Signature of Internship
Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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DECLARATION

I, **PAVITHRA R**, final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **SUBEX LIMITED**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI038

Place: Bangalore

NAME: PAVITHRA R

OFFER LETTER



☎	+91 80 3745 1377
✉	info@subex.com
🌐	www.subex.com

24-Jan-2024

Pavitra R

Dear Pavitra,

Further to our discussions, we are pleased to offer you the position of **Intern**, a temporary position currently available in our organization, Subex Limited based at Bangalore with effect from **08-Feb-2024** under the following terms and conditions.

1. The initial duration will be for **6 Months** effective **08-Feb-2024**.
2. You are entitled to an all-inclusive monthly employment of **INR 25,000** and you will not be entitled for the benefits available for the permanent employees of the Firm. The Firm shall have no liability or responsibility for any, collection, payment or remittance of income taxes, employment insurance, Employer Health Tax, Pension Plan contributions, or payments of any similar or other nature or kind on behalf of or in respect of or for the benefit of you, except for any statutory withholdings and other taxes. You agree to indemnify and hold the Firm harmless from and against all costs and expenses incurred and any order, penalty, interest, premium, contribution, tax or amount that may be charged to or assessed or levied against the Firm on behalf of or in respect of you or arising out of payment by the Firm to you of any applicable charge hereunder.
3. You shall be expected to be available for work during all reasonable time, Monday to Friday. You must always comply with the Firm's policies and procedures. You will not disclose any of confidential information of the organization at any point of time and abide by Confidentiality and Non-Solicit Agreement given in Annexure B.
4. This agreement shall commence on the Effective Date and shall terminate as of the earlier of:
 - a. Upon 30 (Thirty) days' advance notice from either the Firm or the Consultant, without cause.
 - b. The exercise of the Firm's right to terminate this Agreement pursuant to clause (a) hereof shall not abrogate the rights and remedies of the terminating party in respect of the breach giving rise to such termination.
 - c. For purposes here of the services of the Consultant may be terminated.
 - d. By the Firm, immediately, without notice if the Consultant: (1) neglects, fail to carry out or unreasonably refuses to perform the duties assigned to the Consultant under or pursuant to this Agreement (2) materially breaches a term, provision, or condition of this Agreement (3) commits an act of fraud, dishonesty, or moral turpitude; or (4) acts in a manner which results in material loss, damage, or injury to the Firm.
5. Your primary role will be in accordance with Annexure A to this agreement, The Firm reserves the right to vary the conditions of your position from time to time.
6. Please note, that your temporary status in the Firm shall not entitle you to any special consideration for permanent or full-time employment, after your completion of the contract term as per clause 1 of this Agreement.
7. If the above terms are agreeable to you, kindly sign two copies of this letter in the spaces provided below and give one executed copy to us.
8. Refer **Annexure-A** for leave entitlement and **Annexure-B** for Confidentiality Agreement.

For Subex Limited

Aziz Kasem Mazumder
Associate Director – Talent Acquisition

Accepted by Pavitra R

Registered Address : Pritech Park - SEZ, Block-09, 4th Floor B Wing
Survey No. 51 to 64/4, Outer Ring Road, Bellandur Village, Varthur Hobli, Bengaluru - 560 103, India

COMPLETION CERTIFICATE



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🌐	www.subex.com

April 17, 2024

Pavithra R,
301578

Sub: Internship Continuation Letter

Dear Pavithra R,

This is to inform you that your internship effective start date from **February 08, 2024** is **under progress**.

Kindly complete your internship tenure in order to provide a completion certificate.

For Subex Limited,

A handwritten signature in blue ink, appearing to be 'Dharendra Kumar Panda'.

Dhirendra Kumar Panda
Associate Director- C&B • Human Resources

Pritech Park - SEZ, Block-09, 4th Floor B Wing
Survey No. 51 to 64/4, Outer Ring Road, Bellandur Village, Varthur Hobli, Bengaluru - 560103. India

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Upendra K B**, Wharf Street Strategies, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

PAVITHRA R [1DS20AI038]

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CHAPTER 1

INTRODUCTION

Subex is a company that specializes in providing software solutions to the telecommunications industry. They offer products and services related to revenue assurance, fraud management, data analytics, and cybersecurity for telecom operators worldwide. I received training in the telecom domain to gain a comprehensive understanding of Subex's products such as ROC, Hypersense, Diamond, and others. This training equips me with the knowledge to understand the functionalities and operations of these products within the telecom industry. I even had hands-on on the product Hypersense. Then I was put into a Project called Project X.

ROC and Hypersense are two products in Subex. ROC is Revenue Operations Centre which is used for Fraud Management, Partner Settlement, Revenue Assurance. Hypersense is also a Revenue Operations Center. While ROC focuses on specific areas, Hypersense offers a broader range of analytics capabilities. There are many customers for ROC and they don't want to shift to Hypersense. So Project X is making ROC accessible through Hypersense.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Subex Limited is a globally recognized provider of business and operations support systems (B/OSS) solutions tailored specifically for the telecommunications industry. Established in 1992, Subex has amassed extensive expertise in addressing the diverse operational challenges faced by communications service providers (CSPs) worldwide. At the core of Subex's offerings are its innovative software products and solutions, meticulously designed to optimize CSP operations, mitigate risks, and maximize revenue streams. With a comprehensive suite of solutions spanning revenue assurance, fraud management, digital trust, network analytics, and IoT security, Subex empowers CSPs to stay ahead in an increasingly competitive and dynamic market landscape.

Subex's suite of revenue assurance solutions enables CSPs to identify and rectify revenue leakages caused by fraud, billing errors, and system inefficiencies, thereby bolstering revenue streams and enhancing financial performance. Complementing its revenue assurance offerings, Subex provides robust fraud management systems equipped with advanced detection algorithms, enabling CSPs to detect, prevent, and respond to various forms of fraudulent activities, safeguarding their assets and ensuring customer trust. Moreover, Subex's focus on digital trust encompasses cybersecurity solutions designed to fortify digital infrastructure and mitigate risks associated with digital transformation initiatives, positioning CSPs to navigate the evolving threat landscape with confidence and resilience.

In addition to its core offerings, Subex delivers cutting-edge network analytics tools that harness the power of data-driven insights to optimize network performance, enhance capacity planning, and ensure superior quality of service (QoS) for end-users.

CHAPTER 3

SCOPE OF WORK

During my internship at Subex Limited, I actively participated in a diverse array of tasks aimed at enhancing my understanding of product, particularly focusing on securities and vulnerabilities within spark-server folders. I delved into the intricate mechanisms of identifying and mitigating security threats, gaining hands-on experience and implementing effective countermeasures. Through practical exercises and mentorship, I honed my skills in employing industry-standard tools and techniques to fortify network defenses and ensure the integrity and confidentiality of sensitive data.

In addition to my exploration of securities and vulnerabilities, I had the opportunity to delve into the intricacies of ARC systems, which play a pivotal role in ensuring the accurate rating and charging of telecommunications services. My responsibilities included handling out-of-sequence scenarios within ARC systems, where I learned to troubleshoot issues, reconcile discrepancies, and optimize system performance to uphold billing accuracy and customer satisfaction. Engaging with ARC systems provided me with a comprehensive understanding of the complexities involved in rating and charging processes. Moreover, it afforded me the chance to collaborate with cross-functional teams, enhance my problem-solving skills, and contribute to the seamless delivery of telecom services, reinforcing my passion for the intersection of technology and business operations within the telecommunications industry.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

- **Mastery of Product Engineering Fundamentals**

As a Product Engineering Intern at Subex Limited, I had the privilege of immersing myself in the dynamic world of telecommunications software development, where innovation and precision intersected to shape the future of digital connectivity. Throughout my tenure, I engaged in a spectrum of responsibilities aimed at refining product features, optimizing performance, and advancing the overall quality of Subex's offerings.

- **Technical Skill Enhancement**

Enhance your proficiency in programming languages, frameworks, and tools relevant to telecommunications software development, such as Java, Python, Spring Boot, Angular, and Docker. Develop hands-on experience with database management systems, cloud computing platforms, and DevOps practices to optimize product performance and scalability.

- **Domain Knowledge Acquisition**

Deepen your understanding of the telecommunications industry, including its regulatory landscape, market trends, and emerging technologies. Explore the unique challenges and opportunities faced by communications service providers (CSPs) and gain insights into customer requirements, pain points, and competitive dynamics.

- **Communication and Collaboration**

Interns Cultivate effective communication and collaboration skills by engaging with cross-functional teams, stakeholders, and clients. Learn to articulate technical concepts clearly, participate in brainstorming sessions, and contribute to collaborative decision-making processes to drive product innovation and excellence.

- Quality Assurance and Testing

Learn to implement robust testing strategies and quality assurance processes to ensure the reliability, security, and performance of Subex's products. Gain proficiency in automated testing frameworks, continuous integration/continuous deployment (CI/CD) pipelines, and defect tracking systems to drive continuous improvement and minimize software defects.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Navigating the telecommunications domain and comprehending the intricacies of telecom products presented formidable challenges for me. The sheer complexity of telecom systems, coupled with the rapid pace of technological advancement within the industry, initially felt overwhelming. Deciphering the myriad telecom products and services offered by Subex Limited, required a significant investment of time and effort. From the intricacies of network protocols to the complexities of billing systems and service provisioning, each facet of the telecom landscape seemed to present its own set of challenges. Moreover, the dynamic nature of the industry meant that new technologies and market trends were constantly emerging, further adding to the complexity. However, I approached these challenges with determination and a willingness to learn. Through dedicated study, hands-on experience, and collaboration with colleagues and mentors, I gradually gained a deeper understanding of the telecom domain and the products and services that underpin it. Over time, what initially seemed daunting transformed into an area of expertise, enabling me to contribute effectively to product engineering initiatives and navigate the telecom landscape with confidence.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Securities and Vulnerabilities

Apache Maven, renowned as a pivotal build automation tool, predominantly caters to Java projects, streamlining the intricate process of project management and dependency resolution. At its core, Maven harnesses the power of a project object model (POM) file, serving as a blueprint to delineate the project's configuration parameters and its requisite dependencies. Our task entails the generation of a comprehensive Maven dependency tree, meticulously unraveling both direct and indirect dependencies. Through meticulous dependency analysis, we aim to discern the intricate web of dependencies, shedding light on the hierarchical structure and dependencies' interrelationships. Following this, our focus shifts towards the generation of a meticulously crafted report, delineating the direct and indirect dependencies. This report serves as a cornerstone for our subsequent endeavors, offering invaluable insights into the project's dependency landscape and facilitating informed decision-making.

Moreover, our objectives extend beyond mere dependency analysis, delving into the realm of security vulnerability assessment. Leveraging Maven's capabilities, we embark on the journey to unearth potential vulnerabilities lurking within our project's dependencies. These vulnerabilities, categorized based on severity into Critical, High, Medium, and Low, are meticulously documented in a comprehensive report. This report not only serves as a testament to our commitment to robust security practices but also empowers us with the necessary insights to proactively address and mitigate potential security risks. Armed with this invaluable information, we embark on the journey to remediate these vulnerabilities, initiating a systematic process of patching, upgrading, or replacing vulnerable dependencies. Through this proactive approach to vulnerability management, we strive to fortify our project's security posture and safeguard against potential threats, thereby ensuring the resilience and integrity of our software ecosystem.

ARC - Handling Out of Sequences

At the heart of our project lies the pivotal functionality of Accurate Rating and Charging (ARC), which stands as the cornerstone for ensuring precision and reliability in processing Call Detail Records (CDRs). Our primary objective revolves around delivering seamless and precise rating and charging capabilities for these CDRs, a mission critical aspect of telecommunications operations. To achieve this, our focus has been on refining the process of sorting CDRs based on their start date and time, a fundamental step in the accurate determination of call charges and billing cycles. Through meticulous scripting and algorithmic enhancements, we endeavor to optimize the sorting mechanism, ensuring that CDRs are arranged in chronological order, thereby facilitating streamlined and accurate rating and charging processes.

Furthermore, a key facet of our ongoing efforts involves enhancing the efficiency and timeliness of our output file generation. Recognizing the importance of real-time processing and rapid response times, we are in the process of modifying our script files to ensure that each output file is produced at intervals as precise as 5 milliseconds. This ambitious endeavor necessitates intricate adjustments to our scripting logic and file handling mechanisms, as we strive to strike the delicate balance between performance optimization and resource utilization. While this task remains a work in progress, our dedication to precision and efficiency drives us forward, propelling us towards the realization of our goal of providing unparalleled accuracy and responsiveness in the realm of rating and charging for telecommunications services. Through ongoing collaboration and relentless pursuit of excellence, we remain committed to pushing the boundaries of innovation and delivering tangible value to our stakeholders in the ever-evolving landscape of telecom operations.

Frontend in Angular

Embarking on the journey of frontend development with Angular involves immersing oneself in a comprehensive learning process aimed at mastering this powerful framework for building modern web applications. Angular, maintained by Google, has emerged as a leading choice for frontend development due to its robust features, scalability, and extensive ecosystem of tools and libraries. As I delve into learning Angular, I am confronted with a rich array of concepts and techniques that form the foundation of frontend development. From understanding the basics of TypeScript, the language of choice for Angular development, to grasping the intricacies of Angular's architecture and component-based structure, every step in this learning journey unveils new insights and opportunities for growth.

At the core of Angular lies the concept of components, which serve as the building blocks of Angular applications. Learning to create and manage components, along with mastering concepts such as data binding, event handling, and template syntax, lays the groundwork for crafting dynamic and interactive user interfaces. Moreover, delving into Angular's powerful features such as dependency injection, routing, and form handling equips me with the tools necessary to develop robust and maintainable frontend applications. Through hands-on exercises, tutorials, and project-based learning, I not only gain proficiency in Angular's core concepts but also cultivate a deeper understanding of best practices and architectural patterns prevalent in modern frontend development.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on my internship experience as a Product Engineer at Subex Limited, I am struck by the wealth of knowledge gained, challenges overcome, and growth achieved throughout the journey. Working within a dynamic and innovative environment like Subex provided me with invaluable opportunities to apply theoretical knowledge to real-world scenarios, collaborate with talented professionals, and contribute meaningfully to the development of cutting-edge telecommunications solutions.

One of the most rewarding aspects of my internship was the hands-on experience gained in product engineering processes. From conceptualization to implementation and testing, I had the chance to participate in various stages of the product development lifecycle. This allowed me to deepen my understanding of software engineering principles, agile methodologies, and industry best practices. Through mentorship and guidance from experienced engineers, I honed my technical skills, expanded my proficiency in programming languages and frameworks, and gained insights into the complexities of building robust and scalable software solutions tailored to the telecom domain.

Moreover, my internship at Subex provided me with exposure to a diverse range of technologies and tools used in telecom product development. Whether it was learning about Apache Maven for dependency management, exploring Angular for frontend development, or delving into cybersecurity practices for ensuring product integrity, each experience broadened my skill set and enriched my understanding of the intricate interplay between technology and business within the telecom industry.

CHAPTER 8

RECOMMENDATIONS

I suggest exploring interdisciplinary collaborations with experts in fields such as cybersecurity and network engineering to gain deeper insights into the unique challenges faced by CSPs. By integrating diverse perspectives and expertise, Subex can develop holistic solutions that address the evolving needs of its clients and stay ahead of industry trends. Additionally, fostering a culture of diversity and inclusion within the organization can stimulate creativity and innovation, as employees from diverse backgrounds bring fresh perspectives and ideas to the table.

Moreover, I encourage the adoption of best practices in data governance and ethics to ensure the responsible and ethical use of data in all our projects. As Subex continues to innovate and expand its offerings, it is essential to prioritize transparency, accountability, and fairness in data-driven initiatives. By upholding the highest standards of integrity and professionalism, Subex can build trust with its stakeholders and establish itself as a leader in ethical data management practices within the telecommunications industry.

CHAPTER 9

CONCLUSION

My internship experience at Subex provided me with an immersive learning environment that transcended the conventional boundaries of academic study. Within the dynamic realm of telecommunications, I found myself surrounded by a wealth of knowledge and expertise, offering me a gateway to dive deep into the intricacies of this ever-evolving industry. From the outset, I was captivated by the complexity and interconnectedness of telecommunications systems, realizing that my internship would be an enriching journey of discovery and growth.

Subex's commitment to fostering a culture of learning and innovation became evident from the moment I stepped through its doors. Through structured training programs, mentorship opportunities, and hands-on projects, I was equipped with the tools and resources necessary to navigate the multifaceted landscape of the telecom industry with confidence and competence. Each day presented new challenges and learning opportunities, pushing me to expand my horizons, question assumptions, and push the boundaries of my knowledge.

As I delved deeper into my internship at Subex, I found myself immersed in a world where theoretical concepts seamlessly melded with practical application. Whether it was understanding the intricacies of revenue assurance, unraveling the complexities of fraud management, or delving into the nuances of network analytics, every aspect of my internship experience was a testament to Subex's dedication to providing a holistic learning experience. Through hands-on projects, collaborative discussions, and exposure to cutting-edge technologies, I gained invaluable insights into the inner workings of the telecom industry and the critical role that companies like Subex play in shaping its trajectory.

Moreover, beyond the technical aspects of my internship, I was also profoundly impacted by the culture of innovation and collaboration that permeated every facet of Subex's operations. The emphasis on teamwork, creativity, and continuous improvement fostered an environment where ideas flourished, boundaries were challenged, and breakthroughs were celebrated. This culture of innovation not only inspired me to strive for excellence in my work but also instilled in me a sense of purpose and belonging within the Subex community.

CHAPTER 10

REFERENCES

- [1] <https://www.subex.com/>
- [2] <https://www.linkedin.com/company/subex/>
- [3] <https://twitter.com/subex>
- [4] <https://www.facebook.com/SubexLimited/>